Car Lift



Category: **Automotive** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Car lift -4Ton

What it is used for

A four-tonne car lift raises and supports a vehicle for inspection, maintenance, or repair. Correct vehicle positioning, rated-capacity checks, lifting-point selection, and mechanical locking are essential to prevent vehicle falls, crushing injuries, and equipment damage.

Main parts

- Lift posts or supporting structure
- Lift arms and adjustable vehicle pads
- Hydraulic cylinders or drive mechanism
- Power unit and operating controls
- Mechanical safety locks and release mechanism
- Vehicle contact pads, anchors, and emergency stop, if fitted

Before you start

1. Confirm that the lift is rated for the vehicle's weight, dimensions, and centre-of-gravity requirements; do not exceed the four-tonne rating.
2. Inspect the posts, arms, pads, cables or hydraulic lines, anchors, safety locks, and controls for damage or wear.
3. Verify that the lift is installed on an approved foundation and that required inspections are current.
4. Check the vehicle manufacturer's lifting points and ensure the floor is level, clean, and free of obstacles.
5. Confirm that doors, roof accessories, and other vehicle features have adequate clearance during lifting.
6. Keep unauthorised people away, wear safety footwear, and ensure the lift operating area is clearly visible.

How to use

1. Drive the vehicle slowly into the lift bay and align it centrally between the posts.
2. Apply the parking brake, select the recommended transmission position, and switch off the engine as required by the procedure.
3. Position the lift arms and pads at the vehicle manufacturer's approved lifting points.
4. Adjust each arm so the pads make secure contact without contacting fragile components or body panels.
5. Check that all arms and pads are locked in position and that the vehicle is correctly balanced.

6. Clear people, tools, and obstructions from under and around the vehicle.
7. Operate the lift to raise the vehicle a short distance and stop to recheck pad contact and vehicle stability.
8. Continue lifting smoothly while observing the vehicle, arms, and lift for movement or abnormal noise.
9. Raise the vehicle to the required height and lower it gently onto the mechanical safety locks according to the lift design.
10. Confirm that the safety locks are engaged before working beneath the vehicle; never rely on hydraulic pressure alone.
11. Before lowering, remove tools and personnel from the danger area and raise the lift slightly if required to release the locks.
12. Disengage the safety locks using the approved release procedure and lower the vehicle smoothly to the ground.
13. Confirm that all pads are clear, retract the arms, and move the vehicle out of the bay when safe.
14. Report any abnormal operation and record inspections or maintenance concerns.

⚠ Safety

- Only trained and authorised personnel should operate the car lift.
- Never exceed the lift's rated capacity or lift a vehicle with unknown weight distribution without an approved assessment.
- Use only the vehicle manufacturer's approved lifting points and the lift manufacturer's operating procedure.
- Never work under a raised vehicle until the mechanical safety locks are engaged and stability is verified.
- Keep people clear of the vehicle and moving arms during lifting and lowering.
- Do not use a lift with damaged anchors, arms, pads, cables, hydraulic lines, or safety locks; isolate it and report the defect.

Care & storage

- Inspect lift arms, pads, anchors, safety locks, cables, and hydraulic lines at the specified intervals.
- Check hydraulic fluid and lubricate designated moving points according to the manufacturer's instructions.
- Keep the lift bay and floor clean, dry, and free of tools or debris.
- Test safety locks, controls, and emergency stop according to the maintenance schedule.
- Arrange qualified inspection and repair for leaks, uneven lifting, abnormal movement, or damaged components.

Common mistakes to avoid

- Lifting a vehicle above the rated capacity or with incorrect weight distribution.
- Positioning lift pads away from approved vehicle lifting points.
- Working under the vehicle before mechanical locks are engaged.
- Lowering the lift while people, tools, or obstructions remain in the danger area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.